

1. Replacing an inactivated allele for a gene of interest produces a _____ mouse, while replacing a gene with another that has an altered function creates a _____ mouse.
 - a. Knockin, knockout
 - b. Knockin, knockin
 - c. Knockout, knockin
 - d. Knockout, knockout
2. Which of the following Cas proteins is most likely to be used to increase gene expression in animals?
 - a. dCas9
 - b. dCas13
 - c. Cas13
 - d. Cas9
3. Which of the following types of genetic manipulations allow a researcher to experimentally increase gene expression in a mouse model?
 - a. Knockdown
 - b. Transgenic
 - c. Knockoff
 - d. Knockout
4. The following are techniques used for animal transgenesis:
 - I. Retrovirus-mediated transgenics
 - II. Embryonic stem cell method
 - III. Gene guns
 - IV. Sperm-mediated transfer
 - V. Pronuclear microinjectionWhich of the above techniques requires the presence of DNA-binding proteins?
 - a. IV
 - b. III
 - c. V
 - d. II
 - e. I
5. Rats are superior to mice for early drug toxicity tests because:
 - a. Their larger size facilitates surgical and physiological experimentation

- b. They have more human-like responses to drugs
 - c. All of these
 - d. More toxicological data has been collected from mice
6. The following are techniques used for animal transgenesis:
- I. Retrovirus-mediated transgenics
 - II. Embryonic stem cell method
 - III. Gene guns
 - IV. Sperm-mediated transfer
 - V. Pronuclear microinjection
- Which of the above techniques does not require the use of vectors at all?
- a. II
 - b. I
 - c. III
 - d. V
 - e. IV
7. Which of the following statements about monoclonal antibody production is true?
- a. B cell + spleen cell -> hybridoma
 - b. B cell + hybridoma -> myeloma
 - c. T cell + hybridoma -> myeloma
 - d. B cell + myeloma -> hybridoma
8. Which of the following animals are commonly used as bioreactors to produce proteins?
- a. Mice
 - b. Cows
 - c. Goats
 - d. Chickens
9. The following are techniques used for animal transgenesis:
- I. Retrovirus-mediated transgenics
 - II. Embryonic stem cell method
 - III. Gene guns
 - IV. Sperm-mediated transfer
 - V. Pronuclear microinjection
- Which of the above techniques involves the use of linker proteins?
- a. I

- b. V
- c. III
- d. II
- e. IV

10. Production of monoclonal antibodies by hybridoma technology requires the following EXCEPT:

- a. The use of PEG as fusagen
- b. Isolation of cells from the liver
- c. Injection of mice with purified antigen
- d. Isolating single cells into wells

11. Rats are superior to mice for early drug toxicity tests because:

- a. All of these
- b. They have more human-like responses to drugs
- c. More toxicological data has been collected from mice
- d. Their larger size facilitates surgical and physiological experimentation

12. On average, how many years does it take for a drug to be approved for use in humans?

- a. 4
- b. 12
- c. 2
- d. 8

13. Which of the following is not true about Myeloma?

- a. They are cancer cells
- b. They can produce antibodies
- c. They are used in hybridoma technology
- d. They are B lymphocytes

14. The following are techniques used for animal transgenesis:

- I. Retrovirus-mediated transgenics
- II. Embryonic stem cell method
- III. Gene guns
- IV. Sperm-mediated transfer
- V. Pronuclear microinjection

Which of the above techniques involves the use of pluripotent cells?

- a. V
- b. II
- c. III
- d. IV
- e. I

15. Strains of bacteria that are resistant to all antibiotics commonly used today are called:

- a. Competent bacteria
- b. Superbugs
- c. Plaques
- d. Transformants

16. Several lux genes encode protein subunits that form an enzyme called:

- a. Cellulase
- b. Luciferase
- c. Green fluorescent protein
- d. X-galase

17. Which of the following enzymes is used in laundry detergents?

- a. Cellulose
- b. Chymase
- c. Chymotrypsin
- d. Subtilisin

18. The process of weakening a pathogen is called:

- a. Virulence reduction
- b. Vaccination
- c. Immunization
- d. Attenuation

19. What is a potential risk of taking too many antibiotics for infections?

- a. Antibiotics kill bacteria that keep us healthy
- b. Antibiotics “feed” the harmful bacteria
- c. There is no risk in taking antibiotics

- d. The microbiota becomes resistant to antibiotics

Q1	Q2	Q3	Q4	Q5	Q6	Q7	Q8	Q9	Q10	Q11	Q12	Q13	Q14	Q15	Q16	Q17	Q18	Q19
C	A	B	C	C	D	D	C	B	B	A	B	D	B	B	B	D	D	A